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[1]:
```python
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

from sklearn.datasets import load_diabetes
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.linear_model import Ridge
from sklearn.metrics import mean_squared_error, r2_score
data = load_diabetes(as_frame=True)
df = data.frame
print("First 5 rows:")
print(df.head())
print("\nInfo:")
print(df.info())

print("\nMissing Values:")
print(df.isnull().sum())

print("\nSummary:")
print(df.describe())
plt.figure(figsize=(10, 8))
sns.heatmap(df.corr(), cmap="coolwarm")
plt.title("Correlation Heatmap")
plt.show()
X = df.drop("target", axis=1)
y = df["target"]
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42)
scaler = StandardScaler()
X_train_scaled = scaler.fit_transform(X_train)
X_test_scaled = scaler.transform(X_test)
ridge = Ridge(alpha=1.0)
ridge.fit(X_train_scaled, y_train)
y_pred = ridge.predict(X_test_scaled)
print("\nPerformance:")
print("MSE:", mean_squared_error(y_test, y_pred))
print("R2 Score:", r2_score(y_test, y_pred))
coef_df = pd.DataFrame(
    "Feature": X.columns,
    "Coefficient": ridge.coef_
).sort_values(by="Coefficient", ascending=False)

print("\nCoefficients:")
print(coef_df)
plt.figure(figsize=(8, 6))
plt.scatter(y_test, y_pred, alpha=0.5)
plt.xlabel("Actual")
plt.ylabel("Predicted")
plt.title("Actual vs Predicted")

plt.plot([y_test.min(), y_test.max()],
         [y_test.min(), y_test.max()],
         color="red")
```

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plt.plot([y_test.min(), y_test.max()],
         [y_test.min(), y_test.max()],
         color="red")

plt.show()
```

```
First 5 rows:
      age       sex       bmi        bp        s1        s2        s3  \
0  0.038076  0.050680  0.061696  0.021872 -0.044223 -0.034821 -0.043401
1 -0.001882 -0.044642 -0.051474 -0.026328 -0.008449 -0.019163  0.074412
2  0.085299  0.050680  0.044451 -0.005670 -0.045599 -0.034194 -0.032356
3 -0.089063 -0.044642 -0.011595 -0.036656  0.012191  0.024991 -0.036038
4  0.005383 -0.044642 -0.036385  0.021872  0.003935  0.015596  0.008142

         s4        s5        s6  target
0 -0.002592  0.019907 -0.017646   151.0
1 -0.039493 -0.068332 -0.092204    75.0
2 -0.002592  0.002861 -0.025930   141.0
3  0.034309  0.022688 -0.009362   206.0
4 -0.002592 -0.031988 -0.046641   135.0

Info:
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 442 entries, 0 to 441
Data columns (total 11 columns):
 #   Column  Non-Null Count  Dtype
---  ------  --------------  -----
 0   age     442 non-null    float64
 1   sex     442 non-null    float64
 2   bmi     442 non-null    float64
 3   bp      442 non-null    float64
 4   s1      442 non-null    float64
 5   s2      442 non-null    float64
 6   s3      442 non-null    float64
 7   s4      442 non-null    float64
 8   s5      442 non-null    float64
 9   s6      442 non-null    float64
```

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Trusted

JupyterLab | Python [conda env:base] | Anaconda Toolbox

```
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42)
scaler = StandardScaler()
X_train_scaled = scaler.fit_transform(X_train)
X_test_scaled = scaler.transform(X_test)
ridge = Ridge(alpha=1.0)
ridge.fit(X_train_scaled, y_train)
y_pred = ridge.predict(X_test_scaled)
print("\nPerformance:")
print("MSE:", mean_squared_error(y_test, y_pred))
print("R2 Score:", r2_score(y_test, y_pred))
coef_df = pd.DataFrame({
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plt.title("Actual vs Predicted")

plt.plot([y_test.min(), y_test.max()],
         [y_test.min(), y_test.max()],
         color="red")

plt.show()
```

```
Missing Values:
age      0
sex      0
bmi      0
bp       0
s1       0
s2       0
s3       0
s4       0
s5       0
s6       0
target   0
dtype: int64

Summary:
                age           sex           bmi            bp  \
count  4.420000e+02  4.420000e+02  4.420000e+02  4.420000e+02
mean  -2.511817e-19  1.230790e-17 -2.245564e-16 -4.797570e-17
std    4.761905e-02  4.761905e-02  4.761905e-02  4.761905e-02
min   -1.072256e-01 -4.464164e-02 -9.027530e-02 -1.123988e-01
25%   -3.729927e-02 -4.464164e-02 -3.422907e-02 -3.665608e-02
50%    5.383060e-03 -4.464164e-02 -7.283766e-03 -5.670422e-03
75%    3.807591e-02  5.068012e-02  3.124802e-02  3.564179e-02
max    1.107267e-01  5.068012e-02  1.705552e-01  1.320436e-01

                 s1            s2            s3            s4            s5  \
count  4.420000e+02  4.420000e+02  4.420000e+02  4.420000e+02  4.420000e+02
mean   3.918434e-17 -5.777179e-18 -9.042540e-18  9.293722e-17  1.130318e-17
std    4.761905e-02  4.761905e-02  4.761905e-02  4.761905e-02  4.761905e-02
min   -1.156131e-01 -1.023071e-01 -7.639450e-02 -1.260971e-01 -1.377672e-01
25%   -3.035840e-02 -3.511716e-02 -3.949338e-02 -3.324559e-02 -3.317903e-02
```

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JupyterLab | Python [conda env:base] | Anaconda Toolbox

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         color="red")

plt.show()
```
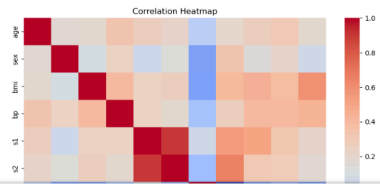

```
           target
count  442.000000
mean   152.133484
std     77.093005
min     25.000000
25%     87.000000
50%    140.500000
75%    211.500000
max    346.000000
```



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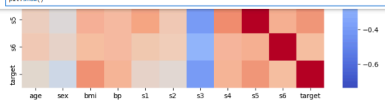


```
Performance:
MSE: 2892.014565758172
R2 Score: 0.4541465207060236

Coefficients:
  Feature  Coefficient
8      s5    31.378384
2     bmi    25.732699
5      s2    17.053075
3      bp    16.734300
7      s4    11.764268
6      s3     3.369914
9      s6     2.458139
0     age     1.807342
1     sex   -11.448190
4      s1   -34.671954
```

Actual vs Predicted

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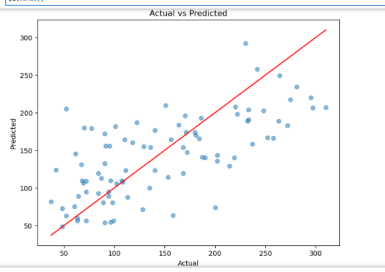

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